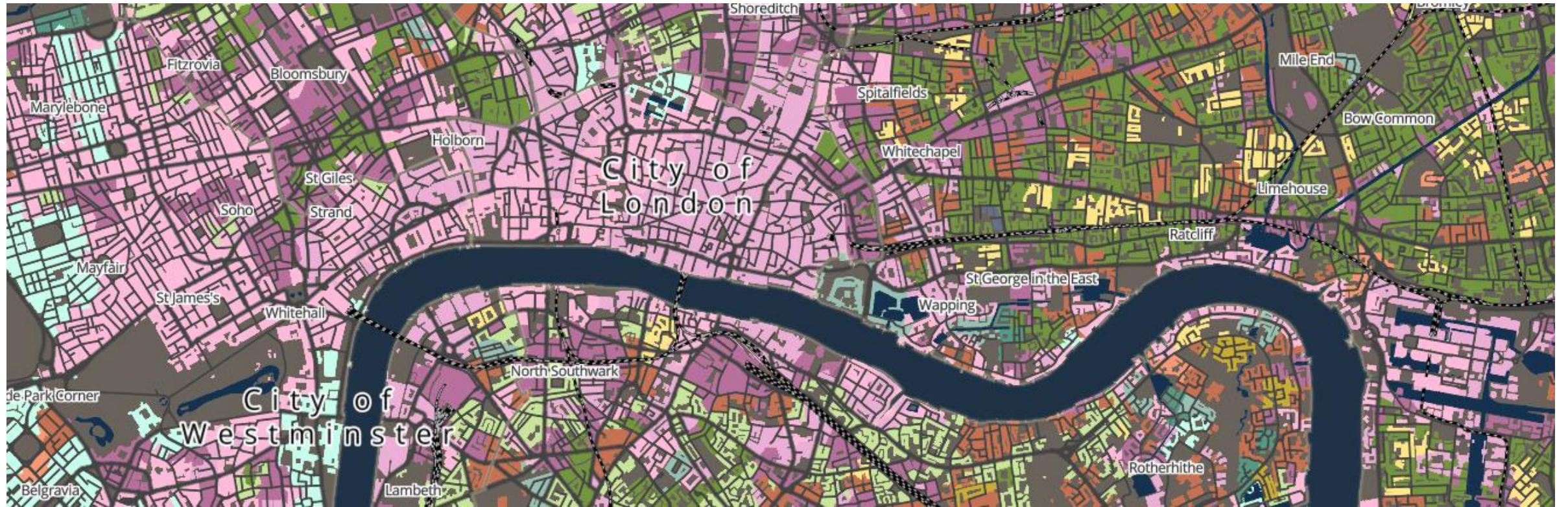


# Geocomputation

## Complex Visualisations



# Module outline

- W1 Reproducible Spatial Analysis
- W2 Spatial Queries and Geometric Operations
- W3 Point Pattern Analysis
- W4 Spatial Autocorrelation
- W5 Spatial Models
- W6 Raster Data Analysis
- W7 Geodemographic Classification
- W8 Accessibility Analysis
- W9 Beyond the Choropleth
- W10 Complex Visualisations

Core Spatial Analysis

Applied Spatial Analysis

Data Visualisation

# This week

- Some pointers for the exam.
- Package and code management.
- Tidy data.
- Grammar of graphics.

Exam



Some pointers for the exam



# Some pointers for the exam

- Two-hour timed exam scheduled for May 28th.
- Two essays (1,000-word limit per question), aim to spend one hour on each.
- Six questions to choose from a range of topics covered in the lectures.
- No questions on coding or programming: focus on theory, underlying principles, methods and applications.
- Q&A Session on TBC.

# How to prepare

- Refer to previous exams to get an idea of the types of questions.
- Strategically read the relevant chapters in the textbook (Longley *et al.* 2018) and more theoretical papers.
- Strategically read the more applied papers, make notes of relevant examples or applications of methods.

# Example question #1

- Critically reflect on the strengths and weaknesses of Kernel Density Estimation in identifying spatial clusters. In your answer make use of Cheshire and Longley (2011).



## Example question #2

- The NHS wants to improve their ambulance response times and they have asked you to assess their current response times in London. Outline the data that you would request, the analytical steps you might take to complete this task and the final maps you could produce in response.

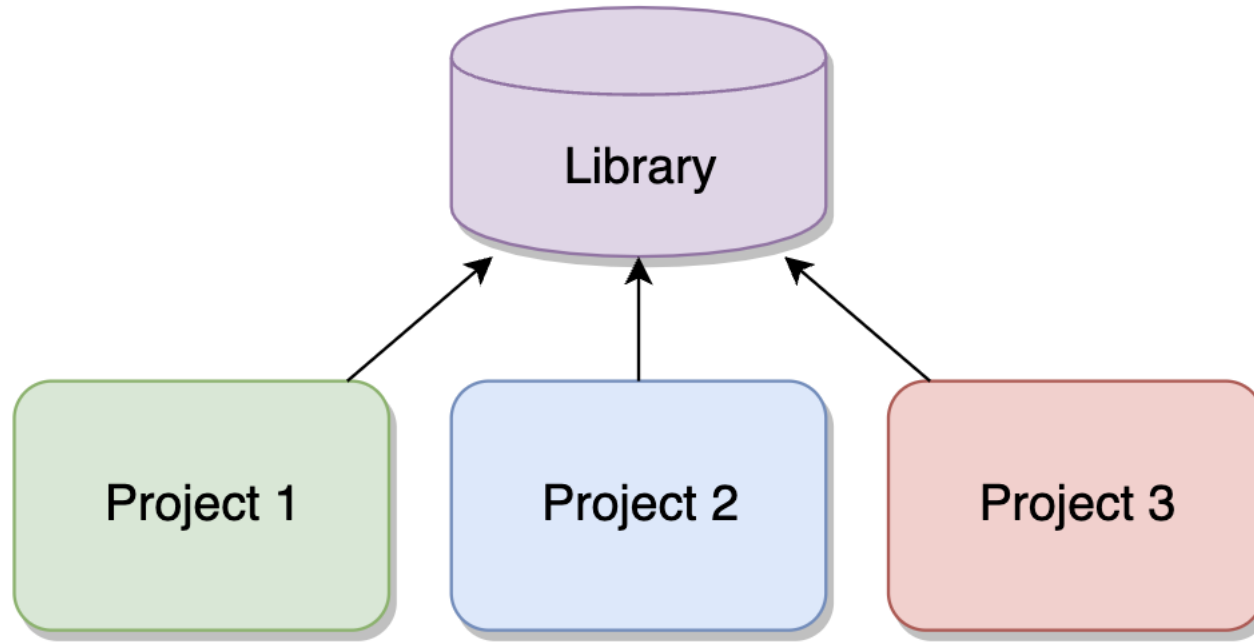
Package and data management

# Package management

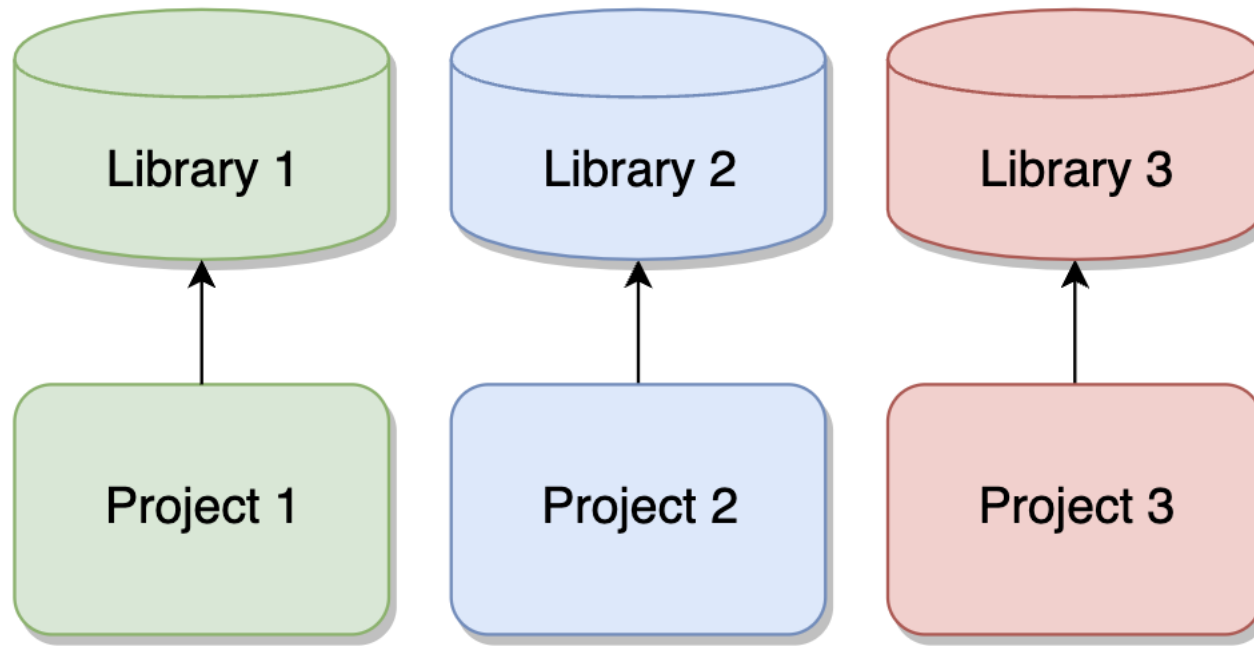
- Package management is the process of handling the many and varied dependencies and artifacts for your servers, applications, and developers.
- Toolkit used to manage project-local libraries.
- Combination with version management through Git.



# Package management



# Package management



# Code management

- Version control: The practice of tracking and managing changes to software code.
- Example: Git (GitHub, GitLab).
- "Track changes" on steroids.
- Beyond the scope of this module but you can still set up your own small version control system ("\_v0.1", "v\_0.2", "v\_1.0").



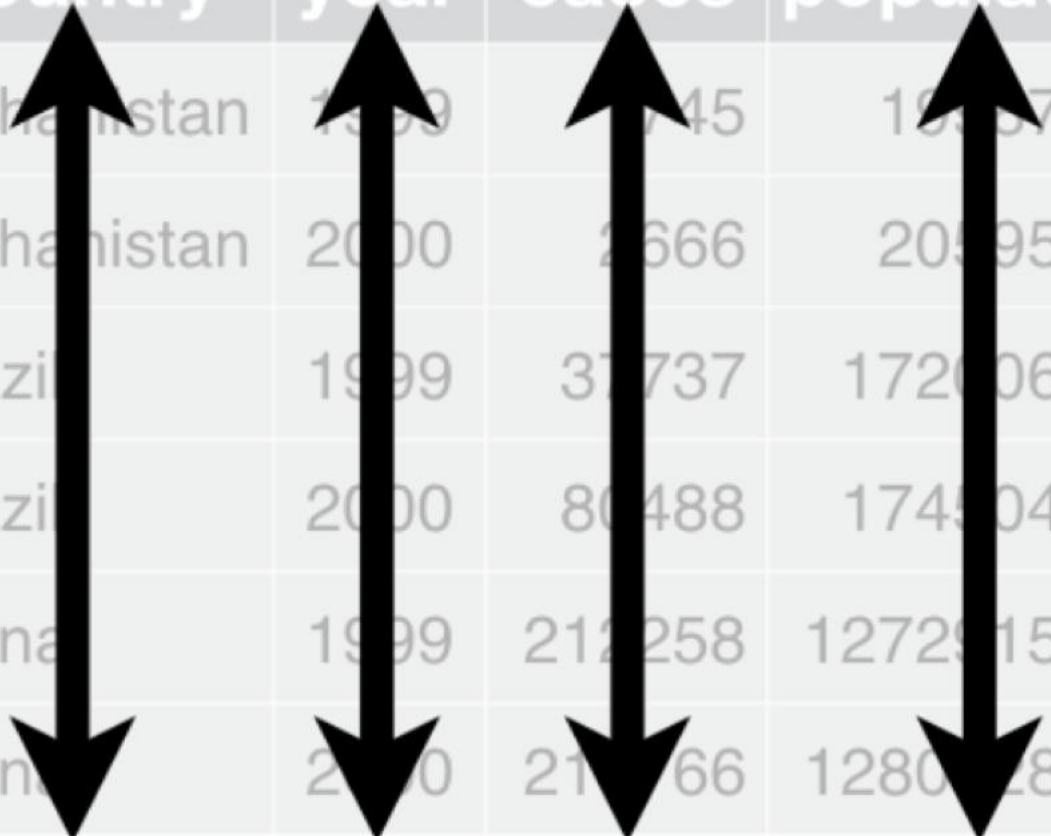
Tidy data

# Managing data

- Wickham 2014. 80 percent of your time goes to data cleaning and preparation ('data wrangling').
- Tidy data refers to the structure and organisation of your data set.
- The idea boils down to three principles.
- Brought together in the tidyverse.

# Tidy data

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19957071   |
| Afghanistan | 2000 | 2666   | 20595360   |
| Brazil      | 1999 | 37737  | 172006362  |
| Brazil      | 2000 | 80488  | 174504898  |
| China       | 1999 | 212258 | 1272915272 |
| China       | 2000 | 21766  | 128028583  |



Each variable must have its own column

# Tidy data

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19987000   |
| Afghanistan | 2000 | 2000   | 20000000   |
| Brazil      | 1999 | 57707  | 172000000  |
| Brazil      | 2000 | 60400  | 174000000  |
| China       | 1999 | 212200 | 127201021  |
| China       | 2000 | 213700 | 128042000  |



Each observation must have its own row

# Tidy data

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19987071   |
| Afghanistan | 2000 | 2666   | 20591360   |
| Brazil      | 1999 | 37737  | 172001362  |
| Brazil      | 2000 | 80488  | 174504898  |
| China       | 1999 | 212258 | 1272915272 |
| China       | 2000 | 213766 | 1280426583 |

Each value must have its own cell

# Tidy data

AutoSave mye22tablesew2023geogsv2.xlsx

Home Insert Draw Page Layout Formulas Data Review View Automate Tell me

Comments Share

Clipboard Font Paragraph Styles Data Tools

General Conditional Formatting Cell Styles Insert Delete Format Sort & Filter Find & Select Sensitivity Analyse Data

A1 MYE2: Persons by single year of age and sex for local authorities in England and Wales, mid-2022

|    | A                                                                                                             | B                          | C                     | D               | E        | F        | G        | H        | I        |
|----|---------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------|-----------------|----------|----------|----------|----------|----------|
| 1  | <b>MYE2: Persons by single year of age and sex for local authorities in England and Wales, mid-2022</b>       |                            |                       |                 |          |          |          |          |          |
| 2  | This worksheet contains one table. Freeze panes are turned on.                                                |                            |                       |                 |          |          |          |          |          |
| 3  | To turn off freeze panes select the 'View' ribbon then 'Freeze Panes' then 'Unfreeze Panes' or use [Alt W, F] |                            |                       |                 |          |          |          |          |          |
| 4  | Please choose from the links presented in the cells below to e-mail us your opinion on this table:            |                            |                       |                 |          |          |          |          |          |
| 5  | <a href="#">This met my needs, please produce it next year</a>                                                |                            |                       |                 |          |          |          |          |          |
| 6  | <a href="#">I need something slightly different (please specify)</a>                                          |                            |                       |                 |          |          |          |          |          |
| 7  | <a href="#">This is not what I need at all (please specify)</a>                                               |                            |                       |                 |          |          |          |          |          |
| 8  | <b>Code</b>                                                                                                   | <b>Name</b>                | <b>Geography</b>      | <b>All ages</b> | <b>0</b> | <b>1</b> | <b>2</b> | <b>3</b> | <b>4</b> |
| 9  | K04000001                                                                                                     | ENGLAND AND WALES          | Country               | 60,238,038      | 625,535  | 621,259  | 645,971  | 658,620  | 668,5    |
| 10 | E92000001                                                                                                     | ENGLAND                    | Country               | 57,106,398      | 596,306  | 592,565  | 615,537  | 627,205  | 635,7    |
| 11 | E12000001                                                                                                     | NORTH EAST                 | Region                | 2,683,040       | 25,453   | 25,572   | 26,261   | 27,574   | 28,1     |
| 12 | E06000047                                                                                                     | County Durham              | Unitary Authority     | 528,127         | 4,649    | 4,696    | 4,876    | 5,020    | 5,1      |
| 13 | E06000005                                                                                                     | Darlington                 | Unitary Authority     | 109,469         | 1,066    | 1,066    | 1,131    | 1,129    | 1,1      |
| 14 | E06000001                                                                                                     | Hartlepool                 | Unitary Authority     | 93,861          | 905      | 941      | 1,021    | 1,040    | 1,0      |
| 15 | E06000002                                                                                                     | Middlesbrough              | Unitary Authority     | 148,285         | 1,775    | 1,683    | 1,764    | 1,857    | 1,8      |
| 16 | E06000057                                                                                                     | Northumberland             | Unitary Authority     | 324,362         | 2,555    | 2,744    | 2,708    | 2,919    | 3,0      |
| 17 | E06000003                                                                                                     | Redcar and Cleveland       | Unitary Authority     | 137,175         | 1,323    | 1,196    | 1,323    | 1,378    | 1,4      |
| 18 | E06000004                                                                                                     | Stockton-on-Tees           | Unitary Authority     | 199,966         | 1,971    | 1,948    | 2,149    | 2,222    | 2,2      |
| 19 | E11000007                                                                                                     | Tyne and Wear (Met County) | Metropolitan County   | 1,141,795       | 11,209   | 11,298   | 11,289   | 12,009   | 12,2     |
| 20 | E08000037                                                                                                     | Gateshead                  | Metropolitan District | 197,722         | 1,921    | 1,942    | 1,958    | 2,036    | 2,0      |
| 21 | E08000021                                                                                                     | Newcastle upon Tyne        | Metropolitan District | 307,565         | 3,107    | 3,084    | 3,070    | 3,230    | 3,2      |
| 22 | E08000022                                                                                                     | North Tyneside             | Metropolitan District | 210,487         | 2,026    | 1,957    | 2,057    | 2,346    | 2,3      |
| 23 | E08000023                                                                                                     | South Tyneside             | Metropolitan District | 148,667         | 1,403    | 1,560    | 1,480    | 1,574    | 1,6      |
| 24 | E08000024                                                                                                     | Sunderland                 | Metropolitan District | 277,354         | 2,752    | 2,755    | 2,724    | 2,823    | 2,9      |
| 25 | E12000002                                                                                                     | NORTH WEST                 | Region                | 7,516,113       | 78,957   | 78,368   | 81,625   | 83,737   | 84,0     |
| 26 | E06000008                                                                                                     | Blackburn with Darwen      | Unitary Authority     | 155,762         | 1,904    | 1,960    | 2,018    | 2,096    | 2,0      |

Correction notice Cover sheet Contents Notes Geography guide Related publications MYE1 MYE2 - Persons MYE2 - Females MYE2 - Males MYE3 MYE4 +

Ready Accessibility: Good to go 150%

# Common errors

- Column headers are values rather than variable names.
- Multiple variables are stored in one column.
- Variables are stored in both rows and columns.
- Multiple observational units are stored in the same column.
- A single observation is stored in multiple tables.

# Tidy ?

| country     | year | type       | count         |
|-------------|------|------------|---------------|
| Afghanistan | 2019 | cases      | 745           |
| Afghanistan | 2019 | population | 19 987 071    |
| Afghanistan | 2020 | cases      | 2 666         |
| Afghanistan | 2020 | population | 20 595 360    |
| Brazil      | 2019 | cases      | 3,7737        |
| Brazil      | 2019 | population | 172 006 362   |
| Brazil      | 2020 | cases      | 80 488        |
| Brazil      | 2020 | population | 174 504 898   |
| China       | 2019 | cases      | 212 258       |
| China       | 2019 | population | 1 272 915 272 |
| China       | 2020 | cases      | 213 766       |
| China       | 2020 | population | 1 280 428 583 |



Tidy ?

| country     | year | rate                    |
|-------------|------|-------------------------|
| Afghanistan | 2019 | 745 / 19,987,071        |
| Afghanistan | 2020 | 2,666 / 20,595,360      |
| Brazil      | 2019 | 3,7737 / 172,006,362    |
| Brazil      | 2020 | 80,488 / 174,504,898    |
| China       | 2019 | 212,258 / 1,272,915,272 |
| China       | 2020 | 213,766 / 1,280,428,583 |

# Tidy ?

## Cases

| country     | 2019    | 2020    |
|-------------|---------|---------|
| Afghanistan | 745     | 2 666   |
| Brazil      | 3,7737  | 80 488  |
| China       | 212 258 | 213 766 |

## Population

| country     | 2019          | 2020          |
|-------------|---------------|---------------|
| Afghanistan | 19 987 071    | 20 595 360    |
| Brazil      | 172 006 362   | 174 504 898   |
| China       | 1 272 915 272 | 1 280 428 583 |

# Tidy ?

| country     | year | cases   | population    |
|-------------|------|---------|---------------|
| Afghanistan | 2019 | 745     | 19 987 071    |
| Afghanistan | 2020 | 2 666   | 20 595 360    |
| Brazil      | 2019 | 3,7737  | 172 006 362   |
| Brazil      | 2020 | 80 488  | 174 504 898   |
| China       | 2019 | 212 258 | 1 272 915 272 |
| China       | 2020 | 213 766 | 1 280 428 583 |

# Grammar of graphics

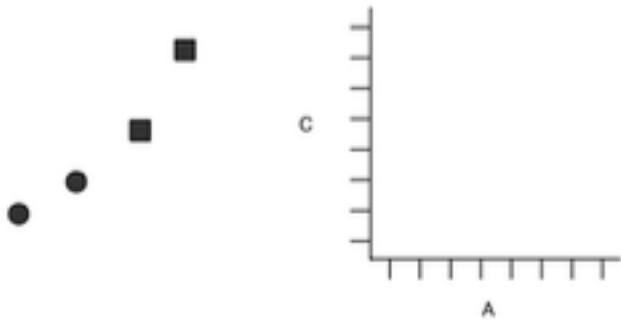
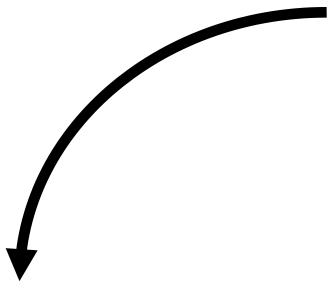
# Grammar of graphics

- Graphics are constructed by layering multiple elements of data.
- Values in a dataset serve as aesthetics: attributes that can be visually represented in a graphic.
- Data, scales, coordinate systems, and plot annotations are layered on top of these aesthetics to create the final graphic.

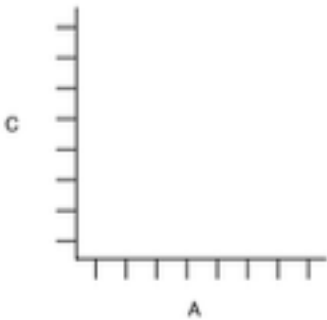
# Grammar of graphics

Dataframe

| <i>x</i> | <i>y</i> | Shape |
|----------|----------|-------|
| 2        | 4        | a     |
| 1        | 1        | a     |
| 4        | 15       | b     |
| 9        | 80       | b     |



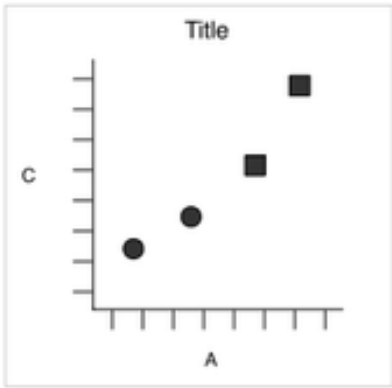
Dataframe  
values



Dataframe  
scale

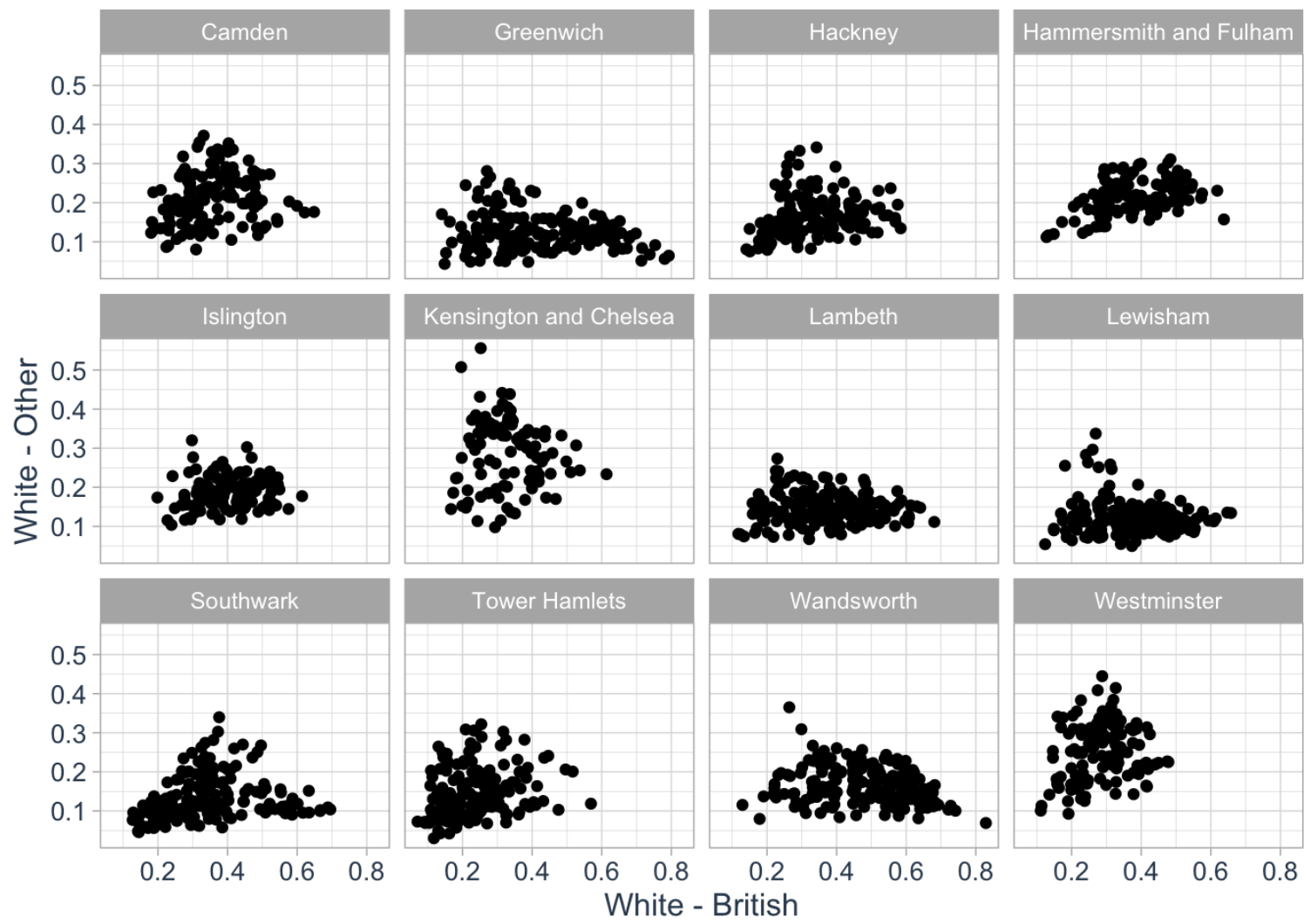


Dataframe  
annotations

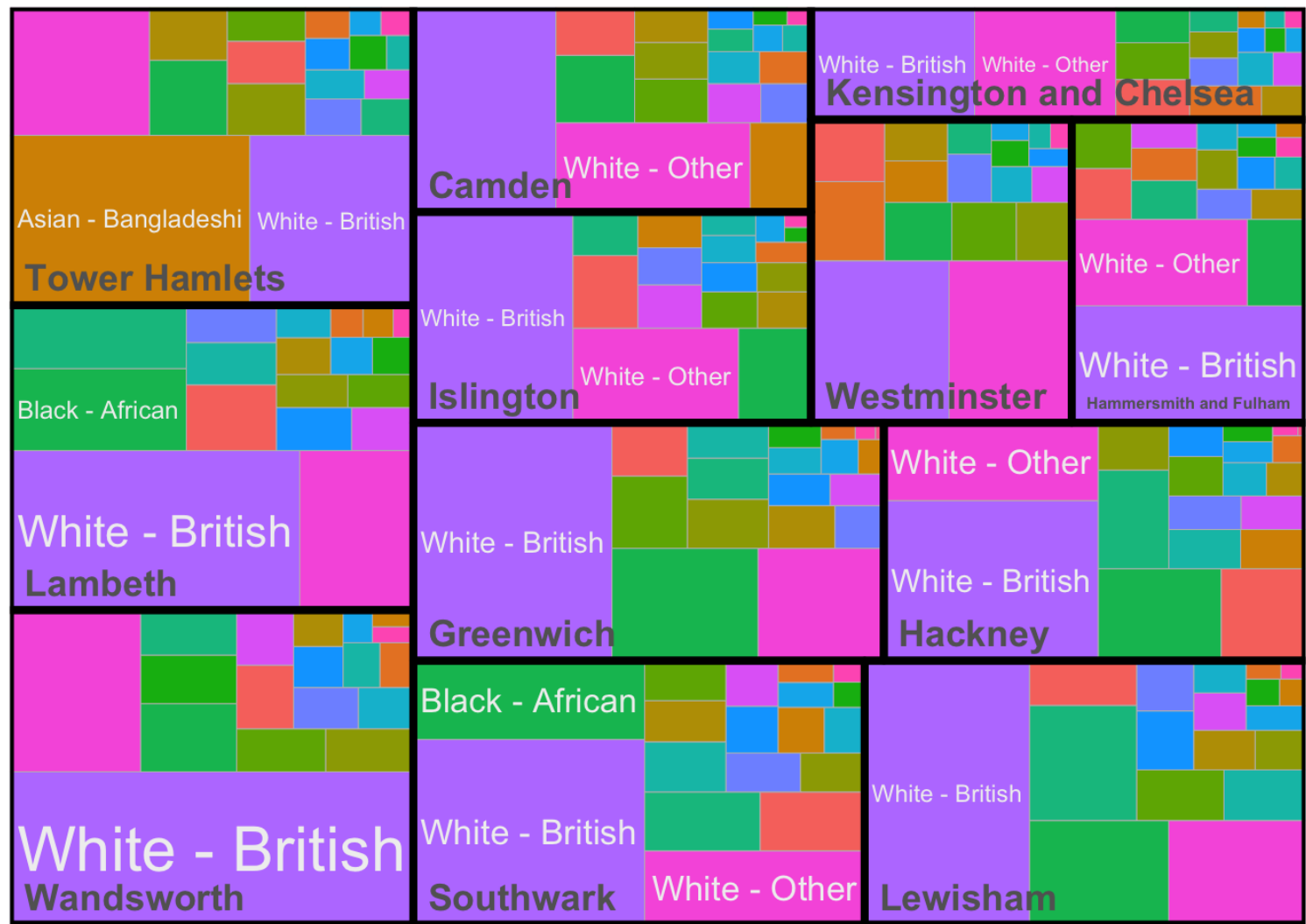


Final  
Graphic

# Grammar of graphics

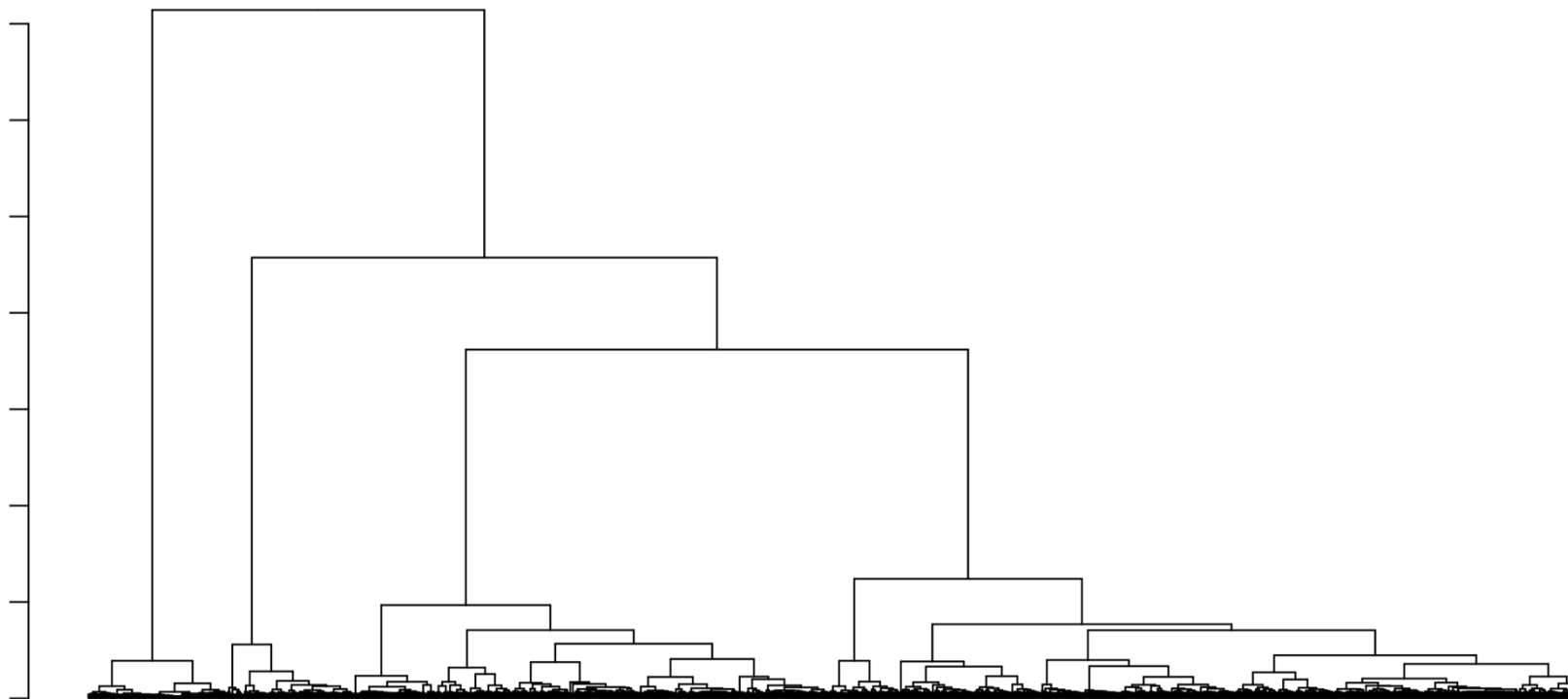


# Grammar of graphics

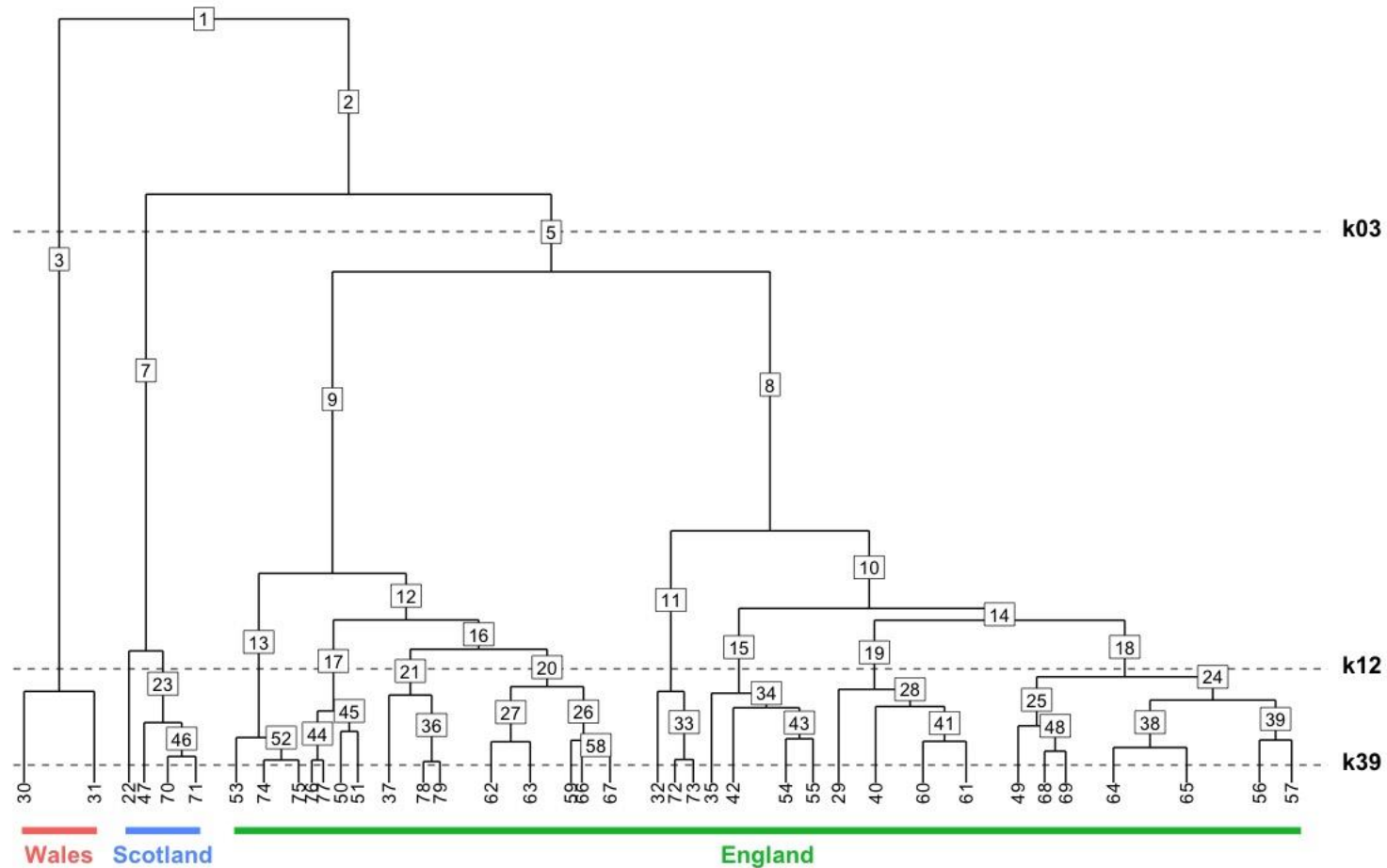




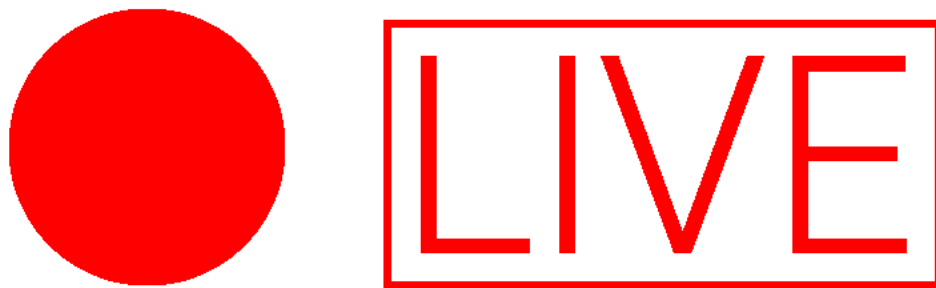
# Grammar of graphics



# Grammar of graphics



RStudio



Everything we covered



# Everything we covered

- 1) Geocomputation as a GIS 2.0: working with geographic data in a computational way, focusing on code, reproducibility and modularity.
- 2) Spatial queries and geometric operations: the core of spatial analysis.
- 3) Working with point event data: special attention to clustering and visualisation of these using DBSCAN and Kernel Density Estimation.
- 4) The First Law of Geography in action: measuring spatial autocorrelation.
- 5) Dealing with spatial autocorrelation and non-stationarity: spatial models

# Everything we covered

- 6) Dealing with raster data: spatial data interpolation.
- 7) Geodemographics: analysis of people by where they live.
- 8) Measuring accessibility: working with a digital network.
- 9) Positioning the map: Coordinate Reference Systems.
- 10) Complex Visualisations: using ggplot2 to build a graphic layer by layer.

# Questions

Justin van Dijk  
[j.t.vandijk@ucl.ac.uk](mailto:j.t.vandijk@ucl.ac.uk)



# Before you go - CMD

- Go to [www.menti.com](https://www.menti.com)
- Use code: 5964 9795



Thank you!

